

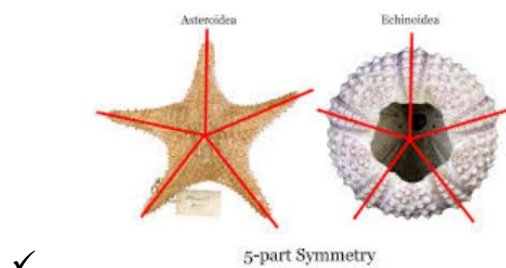
PHYLUM ECHINODERMATA

The echinoderms (**echinos- hedgehog/spinny, derma-skin**) are all marine forms and include sea stars, brittle stars, sea urchins, sea cucumbers and sea lilies.

They comprise of approximately 6500 species.

General characteristics and adaptive features of echinoderms

- ✓ Echinoderms are marine invertebrates known for their unique radial symmetry, allowing their bodies to be divided into five identical parts.
- ✓ Their structure is highly characteristic, with radial symmetry in the adults of most of the representatives and bilateral symmetry in the larvae. In most echinoderms the radial symmetry is pentamerous or star-shaped (have pentaradial symmetry).



- ✓ They are found in the marine environment.
- ✓ **Body Wall:** Possess an endoskeleton composed of calcareous plates called **ossicles** (*small, bony, or spiny calcium carbonate structures that make up the internal endoskeleton of echinoderms*). These may be fused (as in sea urchins) or articulate (as in sea stars). Many have spines protruding through the epidermis.
- ✓ **Body covering and/or skeleton:** Endoskeleton (the name echinoderm means spiny-skinned) of dermal calcareous plates/ossicles with spines or spicules. Body covered with epidermis (often ciliated) sometimes with pedicellariae (*minute, wrench- or pincer-like defensive organs found on the surface of echinoderms, specifically sea stars and sea urchins. These appendages, composed of calcareous ossicles (valves) on stalks, function like scissors to protect the animal by grasping prey removing debris, and defending against predators*).
- ✓ No cephalisation is found and the body can be divided into an oral and an aboral side. The oral side is usually closest to the substrate.
- ✓ Germ layers; Triploblastic coelomates.

- ✓ They have Organ level of organization.
- ✓ They move by means of hydraulic organs, the tube feet as the main locomotory organs. Other types are movement by spines, crawling, swimming and active movement of arms. Some are sessile.
- ✓ Mode of living: All are free living, no parasitic forms occur, some are predators or commensals (*host numerous commensal organisms, including crustaceans, fishes, and polychaete worms, which receive shelter, food, or protection*).
- ✓ **Water Vascular System (WVS):** This is the hallmark of the phylum. It is a network of fluid-filled canals derived from the coelom. It functions in: Locomotion (via tube feet), Respiration, Food and waste transport, Sensory perception.
- ✓ Sexes are nearly always separate, fertilization being external. Development is via radial cleavage, through free-swimming, bilateral larval stages (dipleurula larva) which undergo metamorphosis.
- ✓ The phylum Echinodermata is divided into 5 classes namely: Crinoidea, Asteroidea, Ophiuroidea, Echinodea and Holothuroidea.

Gaseous Exchange Organs in Echinoderms

Echinoderms (such as starfish, sea urchins, and sea cucumbers) lack a centralized respiratory system like lungs or gills found in vertebrates. Instead, they rely on various specialized structures that utilize the **water vascular system** and direct diffusion across thin epithelial membranes.

Since most echinoderms have a fluid-filled body cavity (coelom), gaseous exchange occurs as oxygen diffuses from the surrounding seawater into the coelomic fluid, while carbon dioxide moves out.

1. Dermal Branchiae (Papulae)

- ✓ Found prominently in **Asteroidea** (starfish), these are small, thin-walled outgrowths of the body wall that extend through the ossicles (calcareous plates).
- ✓ **Mechanism:** They are lined with cilia that create a current, bringing fresh oxygenated water into contact with the thin membrane.
- ✓ **Function:** They serve as the primary site for both respiration and the excretion of nitrogenous waste.

2. Tube Feet (Podia)

- ✓ The tube feet are part of the water vascular system and are found in almost all echinoderms.
- ✓ **Mechanism:** Because tube feet have a very thin walls and a large surface area, they act as efficient respiratory surfaces.
- ✓ **Function:** As the animal moves or anchors itself, oxygen diffuses from the water into the fluid within the tube feet and then into the rest of the body.

3. Respiratory Trees

- ✓ Found in sea cucumbers (*Holothuroidea*).
- ✓ Branched, internal structures/branched tubes connected to the cloaca.
- ✓ **Mechanism:** The sea cucumber pumps water in and out of the cloaca (often referred to as "anal breathing"). The water enters the respiratory trees, where oxygen is absorbed into the coelomic fluid before the water is expelled.

4. Peristomial Gills

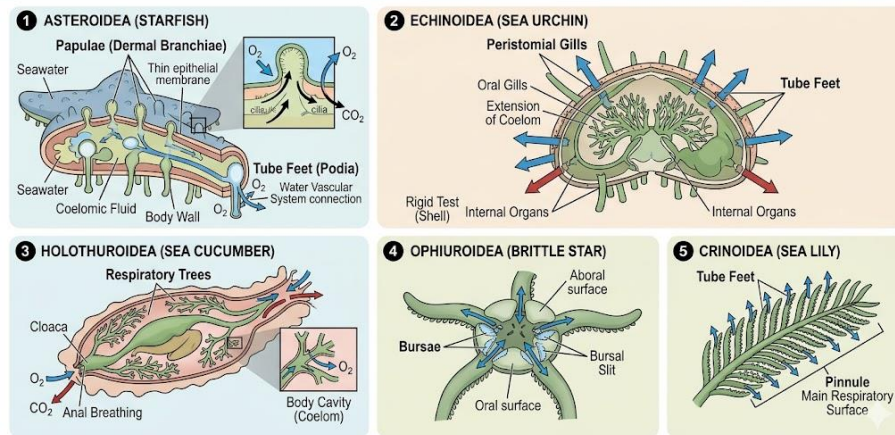
- ✓ In **Echinoidea** (sea urchins), gaseous exchange is supplemented by five pairs of branched gills located around the mouth (peristome).
- ✓ **Mechanism:** These gills are extensions of the coelomic cavity and provide additional surface area for gas exchange, which is necessary because the rigid "test" (shell) of the urchin limits diffusion through the body wall.

5. Bursae (*specialized, ciliated, sac-like invaginations found in echinoderms belonging to the class *Ophiuroidea* (brittle stars and basket stars). These structures are primarily located on the oral side of the central disk, usually appearing as 10 slit-shaped openings situated between the arm bases*)

- ✓ In **Ophiuroidea** (brittle stars), respiration occurs through specialized sacs called bursae.
- ✓ **Structure:** There are typically ten bursae located at the base of the arms on the oral surface.
- ✓ **Mechanism:** Water is circulated in and out of these sacs through narrow slits. Gas exchange occurs across the thin lining of the bursa into the surrounding coelomic fluid.

Class	Common Name	Primary Gaseous Exchange Organ
Astroidea	Starfish	Papulae (Dermal branchiae) & Tube feet
Echinoidea	Sea Urchins	Peristomial gills & Tube feet
Holothuroidea	Sea Cucumbers	Respiratory trees
Ophiuroidea	Brittle Stars	Bursae
Crinoidea	Sea Lilies	Tube feet (Podia)

GASEOUS EXCHANGE ORGANS IN ECHINODERMS



✓

Class	Common Names	Key Features
Asteroidea	Sea Stars	Star-shaped; arms not sharply marked off from central disc; suckered tube feet; gastric eversion for feeding.
Ophiuroidea	Brittle Stars	Long, slender arms sharply marked off from central disc; move by rowing arms; no suckers on tube feet; no anus.
Echinoidea	Sea Urchins, Sand Dollars	No arms; globular or discoidal body; fused ossicles forming a test ; possess Aristotle's Lantern (chewing organ).
Holothuroidea	Sea Cucumbers	Elongated body; leathery skin; reduced ossicles; respiratory trees; mouth surrounded by tentacles. Can eviscerate viscera as defense (it means <i>the animal intentionally expels its own internal organs (viscera) to distract, confuse, or entangle a predator. This extreme defensive action allows the animal to escape, and in many cases, the lost organs are regenerated within a few days or weeks</i>)
Crinoidea	Sea Lilies, Feather Stars	Cup-shaped body (calyx); branched arms with pinnules; filter feeders; many are sessile with a stalk.

Biological Importance

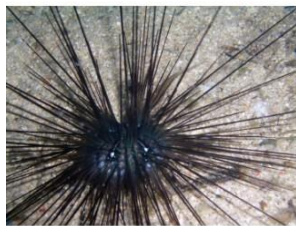
- **Ecological Role:** Many are "keystone species" (e.g., sea stars controlling mussel populations). Sea urchins are major grazers of kelp forests.
- **Blue Economy:** Several species, particularly sea cucumbers (*Holothurians*), are high-value export commodities in the "Beche-de-mer" trade.
- **Research:** Because they are deuterostomes (like chordates), their embryonic development is a primary model for developmental biology.



(a)



(b)



(c)



(d)



(e)

Different members of Echinodermata include the (a) sea star in class Asteroidea, (b) the brittle star in class Ophiuroidea, (c) the sea urchins of class Echinoidea, (d) the sea lilies belonging to class Crinoidea, and (e) sea cucumbers representing class Holothuroidea.

WATER VASCULAR SYSTEM IN ECHINODERMS

It is a unique fluid-based circulatory system composed of a central ring of canals that extend into each arm, powering movement, feeding, gas exchange, and waste removal. Seawater is drawn into this system and harnessed to drive the tube feet, allowing organisms like sea stars and sea urchins to move, grasp prey, and exchange gases.

